

Practice Sheet-1

DSA-1 Lab

Spring 26

Problems

Problem 1. Learn Three Sorting Algorithms in Both Ascending and Descending Order

Implement the following sorting algorithms:

- Bubble Sort
- Insertion Sort
- Selection Sort

For each algorithm, sort the given array in:

- Ascending order
- Descending order

Problem 2. Sort an Array Based on Absolute Value

Given an array of integers, sort the array in non-decreasing order of the absolute values of its elements.

Input:

- An integer n representing the size of the array
- n integers

Output:

- Print the sorted array based on absolute value

Sample Input:

8
-10 3 -2 7 -1 5 -6 2

Sample Output:

-1 -2 2 3 5 -6 7 -10

Problem 3. Sort Only the Even or Odd Index of an Array

Given an array of integers, sort only the elements located at:

- Even indices (keeping odd indices unchanged), or
- Odd indices (keeping even indices unchanged)

while keeping the other positions unchanged.

Input:

- An integer n representing the size of the array
- n integers

Output:

- Print the modified array after sorting only the required indices

Sample Input:

```
7
9 4 7 2 6 1 8
Sort even indices
```

Sample Output:

```
6 4 7 2 8 1 9
```

Problem 4. Sort an Array Based on Frequency of the Elements

Given an array of integers, sort the elements based on their frequency of occurrence.

Input:

- An integer n representing the size of the array
- n integers

Output:

- Print the array sorted based on the frequency of elements

Sample Input:

```
10
4 6 2 4 3 2 2 6 4 5
```

Sample Output:

3 5 6 6 2 2 2 4 4 4

Problem 5. Find the k th Smallest Element in an Array

Given an array of integers and an integer k , find the k th smallest element in the array.

Input:

- An integer n representing the size of the array
- n integers
- An integer k

Output:

- Print the k th smallest element

Sample Input:

8
12 3 7 1 19 5 8 2
4

Sample Output:

5